

# Physics-compiled

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# 1 Classical mechanics

## 1.1 Analytical mechanics: Euler-Lagrange

Analytical mechanics is a total reformulation of Newton's law under the work of many physicists and mathematicians like Lagrange, Maupertuis, Euler, Hamilton and Jacobi. The term "analytical mechanics" was first introduced by Lagrange as the name of his manuscript ("mécanique analytique") in 1788. This work provided a general basis for all modern theories: relativity relies on Lagrangian formulation of mechanics, statistical physics uses Hamilton equations while quantum physics was initially based on Hamilton-Jacobi derivations.

### 1.1.1 Constraints and generalized coordinates

In Newtonian physics, a system of  $N$  point particles in 3d is naturally described in the ambient configuration space  $\mathbb{R}^{3N}$ , with a state given by positions and velocities at a given time,  $s = (\vec{r}_1, \dots, \vec{r}_N, \dot{\vec{r}}_1, \dots, \dot{\vec{r}}_N) \in \mathbb{R}^{6N}$ . While correct, this description is rarely adapted to the geometry of the problem. A student solving a 2d pendulum must project forces onto a constraint direction by hand; a block sliding with friction forces one to introduce normal and frictional forces that are not known in advance and must be solved for jointly with the motion itself. Two weaknesses of the Newtonian approach are visible here:

- Working in the full ambient space  $\mathbb{R}^{3N}$  is often unnecessary and introduces redundant equations when the system is geometrically constrained.
- The forces enforcing these constraints (tension, normal reaction, friction) are frequently unknown *a priori*, and computing them is itself part of the difficulty.

The first conceptual step of analytical mechanics is to stop working in the full ambient space and restrict attention to the coordinates that actually parametrize the accessible configurations [2].

Concretely, many systems are constrained by relations that depend only on positions (and possibly time), called **holonomic constraints**:

$$f_a(\vec{r}_1, \dots, \vec{r}_N, t) = 0, \quad a = 1, \dots, k \quad (1)$$

A pendulum of length  $R$ , for instance, satisfies  $x^2 + y^2 = R^2$ : its motion is confined to a curve, not the full plane  $\mathbb{R}^2$ . A constraint that cannot be reduced to such a position-only relation (typically because it involves velocities in a non-integrable way) is called **non-holonomic**.

Collecting the  $k$  holonomic constraint functions into a single map  $F = (f_1, \dots, f_k) : \mathbb{R}^{3N} \rightarrow \mathbb{R}^k$ , the set of configurations satisfying all constraints simultaneously is exactly the level set:

$$\mathcal{C} = F^{-1}(\{0, \dots, 0\}) \subset \mathbb{R}^{3N} \quad (2)$$

This is precisely the object studied in the section on submanifolds (see mathematical tools notes). When the constraints are independent (i.e., when 0 is a regular value of  $F$  in the sense of the regular value theorem),  $\mathcal{C}$  is a smooth submanifold of  $\mathbb{R}^{3N}$ , called the **configuration space** of the system, of dimension:

$$d = \dim \mathcal{C} = 3N - k \quad (3)$$

This integer  $d$  is the number of **degrees of freedom** of the system. When  $k = 0$ , no constraint is imposed and  $\mathcal{C} = \mathbb{R}^{3N}$ . When  $k = 3N$ , the system is fully immobilized.

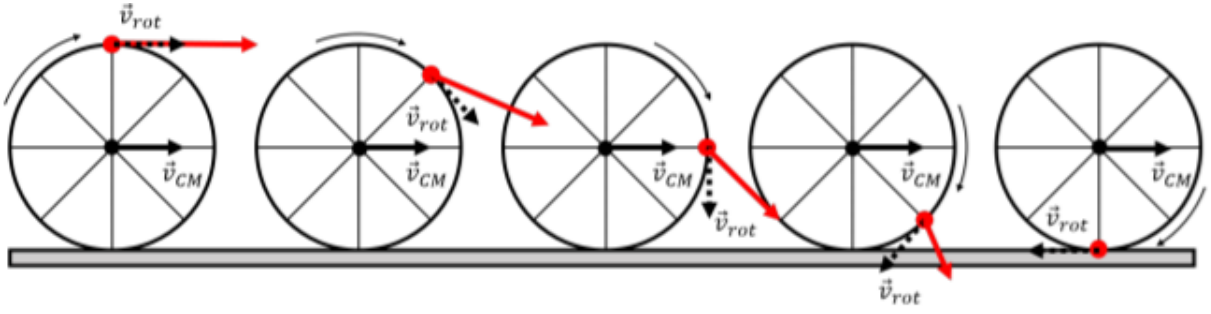


Figure 1: Example of a non-holonomic constraint: when the disk rolls without slipping on the plane, we impose the relative velocity between the disk and the plane to be zero at the contact point. The constraint does not depend only on position and time, it is not holonomic. Taken from [phys.Libretexts.org](https://phys.libretexts.org)

**Example.** A planar pendulum ( $N = 1$ , ambient  $\mathbb{R}^2$ ) has one constraint  $x^2 + y^2 = R^2$ , so  $k = 1$  and  $d = 2 - 1 = 1$ : a single angle suffices to describe the motion.

A holonomic constraint removes an entire direction from  $\mathcal{C}$  itself: the system can never access the forbidden region, at any time. A non-holonomic constraint, by contrast, restricts only which *directions of motion* (which tangent vectors in  $T_p\mathcal{C}$ ) are accessible at each configuration, without removing any configuration from  $\mathcal{C}$  itself. The configuration space therefore stays unreduced, and there is no smaller manifold to which the construction below can be applied directly [4].

A standard example is a disk rolling without slipping on a plane (Fig. 1). The no-slip condition restricts the disk's velocity at the contact point, yet the disk can reach any position  $(x, y)$  and any orientation  $\theta$ : the configuration space remains the full  $\mathbb{R}^2 \times S^1$ , unreduced: the constraint is felt in *how* the disk must move, not in *where* it can go.

In what follows, we restrict ourselves to holonomic constraints unless stated otherwise.

Since  $\mathcal{C}$  is a  $d$ -dimensional smooth submanifold, it admits, near any regular configuration, a local chart  $\varphi : U \subset \mathcal{C} \rightarrow \mathbb{R}^d$ , in the sense developed in the mathematical tools section. The chart's inverse (the local parametrization) expresses the Cartesian positions of all  $N$  particles as smooth functions of  $d$  free parameters  $q^1, \dots, q^d$  (and possibly time, for moving constraints):

$$\vec{r}_j = \vec{r}_j(q^1, \dots, q^d, t), \quad j = 1, \dots, N \quad (4)$$

These parameters are the **generalized coordinates**: by construction, every configuration satisfying the constraints corresponds to some choice of  $(q^1, \dots, q^d)$ , and conversely, with no residual constraint linking them: they are free and independent, exactly the free coordinates of the graph view of a submanifold. The associated velocities  $\dot{q}^\alpha$  are the **generalized velocities**, and the  $2d$ -dimensional space of pairs  $(q^\alpha, \dot{q}^\alpha)$  (the tangent bundle  $T\mathcal{C}$ ) is the reduced state space of the system, generally far smaller than the original  $\mathbb{R}^{6N}$ .

In summary, generalized coordinates are the natural local coordinates of the configuration manifold  $\mathcal{C}$ , adapted to the geometry imposed by the constraints [3].

## 1.1.2 Euler-Lagrange equations

### A. Virtual displacements

Consider a system evolving under holonomic constraints. At a given instant  $t$ , we define a **virtual displacement**  $\delta \vec{r}_i$  as an infinitesimal change of the position of particle  $i$ , taken at *fixed time*, that is compatible with the constraints. In other words, if the system is at a point of  $\mathcal{C}$ , the displaced configuration  $\vec{r}_i + \delta \vec{r}_i$  must also lie in  $\mathcal{C}$ . [1]

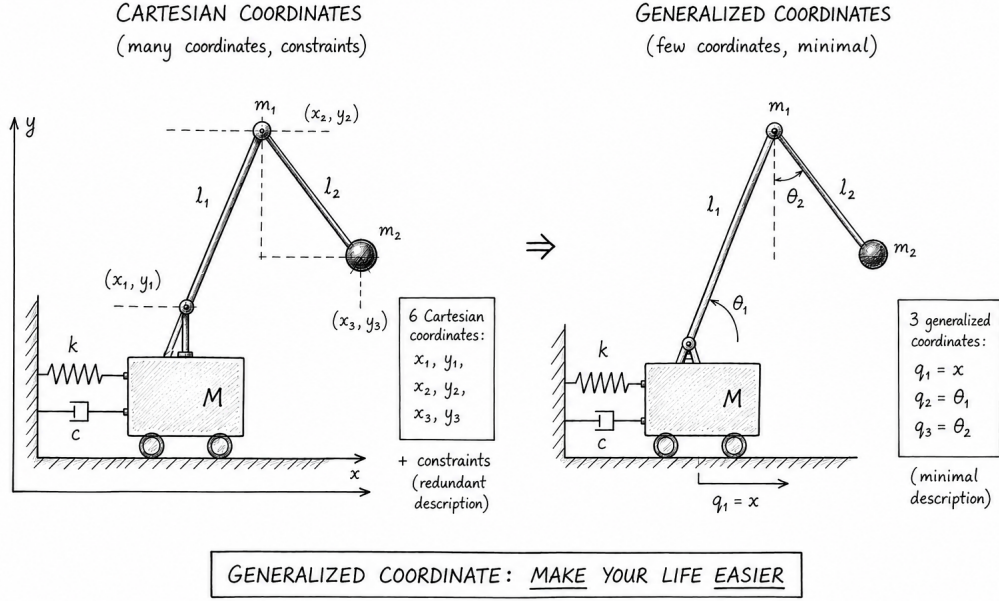


Figure 2: On complicated system such as this, the Cartesian coordinates are fine but will involve a lot of projections, redundancies and reduction of the equation to arrive at a simple form. Meanwhile, working with the generalized coordinates reduces the state space dimension and ensure we are working in the right basis: minimal amount of projections and no redundancies. Illustration generated by AI. Feel free to send me yours to replace it.

This must be contrasted with a **natural displacement**  $d\vec{r}_i = \vec{r}_i(t+dt) - \vec{r}_i(t)$ , which connects two positions on the actual trajectory at two different times, and is governed by the equations of motion. Virtual displacements, by contrast, are purely geometric: they probe the constraint manifold  $\mathcal{C}$  at a frozen instant, with no reference to dynamics.

Since  $\delta$  and  $d$  are derivations acting on different variables (generalized coordinates and time respectively), they commute (by Schwartz theorem):

$$\delta(d\vec{r}_i) = d(\delta\vec{r}_i) \quad (5)$$

This identity has a deeper significance in field theory and quantum mechanics, where it underlies the equivalence of the Lagrangian and Hamiltonian variational principles (see the upcoming notes on quantum mechanics).

In summary, we will remember that a virtual displacement is varying the configuration (system states allowed by the constraints) while freezing the time: it is not a natural displacement, as time did not evolve under this displacement. You can picture it as a remanent image, an "imagined" state.

## B. Virtual work and D'Alembert's principle

The constraint equations introduced earlier are purely mathematical conditions on the positions. Physically, however, something must enforce them: a hand on a table stays on the table because the table exerts some sort of force on it; a pendulum mass stays at fixed distance from the pivot because the rod pulls it back whenever it tries to stray under gravity. We call these **constraint forces**  $\vec{F}_i^c$  the forces that constraint the system on the configuration manifold  $\mathcal{C}$ .

All other forces (gravity, springs and Hooke's law, any interaction prescribed independently of the geometry...) are called **applied forces**  $\vec{F}_i^{\text{app}}$ .

The total force on particle  $i$  is therefore:

$$\vec{F}_i = \vec{F}_i^{\text{app}} + \vec{F}_i^{\text{c}} \quad (6)$$

Constraint forces are typically unknown: we rarely know the tension in a rod or the normal force from a surface before solving the problem. This is precisely the difficulty encountered in Newtonian mechanics. However, their role gives us a crucial geometric property. Since  $\vec{F}_i^{\text{c}}$  must keep the system *on*  $\mathcal{C}$  without driving it *along*  $\mathcal{C}$  (that is the role of an applied force), it can only point in directions orthogonal to  $\mathcal{C}$ . In other words, constraint forces are normal to the constraint manifold. Since virtual displacements are by definition tangent to  $\mathcal{C}$ , constraint forces do no virtual work:

$$\sum_i \vec{F}_i^{\text{c}} \cdot \delta \vec{r}_i = 0 \quad (7)$$

In other words, the virtual work is given only by the applied forces:

$$\delta W = \sum_i \vec{F}_i^{\text{app}} \cdot \delta \vec{r}_i \quad (8)$$

This is not an approximation: it is a direct consequence of the geometry of  $\mathcal{C}$  and of the definition of virtual displacements.

We can now apply Newton's second law  $\vec{F}_i = m_i \ddot{\vec{r}}_i$  to each particle and take the virtual work of both sides. The sum over all particles leads to:

$$\sum_i m_i \ddot{\vec{r}}_i \cdot \delta \vec{r}_i = \sum_i \vec{F}_i^{\text{app}} \cdot \delta \vec{r}_i + \underbrace{\sum_i \vec{F}_i^{\text{c}} \cdot \delta \vec{r}_i}_{=0} \quad (9)$$

This yields to the famous d'Alembert principle:

$$\boxed{\sum_i \left( m_i \ddot{\vec{r}}_i - \vec{F}_i^{\text{app}} \right) \cdot \delta \vec{r}_i = 0} \quad \text{D'ALEMBERT PRINCIPLE} \quad (10)$$

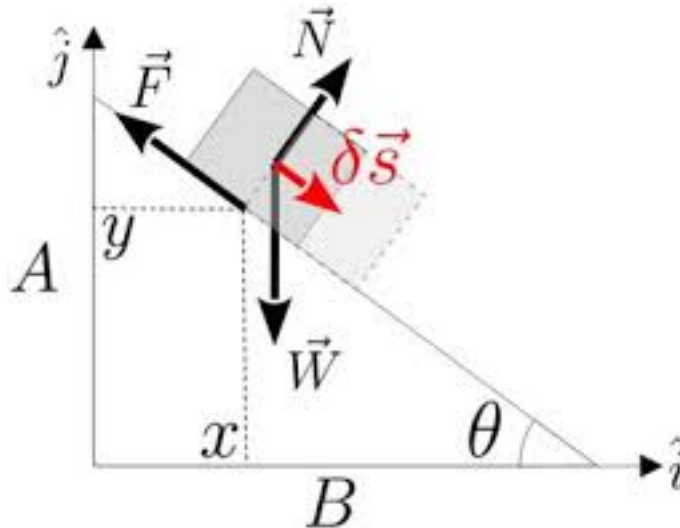


Figure 3: D'Alembert principle states that forces representing constraints are always orthogonal to virtual displacement. In the example of a block pushed along a slope, the normal reaction,  $\vec{N}$ , is indeed always orthogonal to  $\delta \vec{s}$ . Taken from [5].

The constraint forces have disappeared entirely as they do no virtual work. What remains involves only the applied forces (which are known parameters) and the accelerations.

D'Alembert's principle converts Newton's law, which holds in the full ambient space and mixes known and unknown (constraints) forces, into a single scalar equation stated entirely within  $\mathcal{C}$ .

#### remarks

It is crucial to remember that this holds for *holonomic* constraints (smooth surfaces, rigid rods, inextensible strings...). Sliding friction violates this picture: it is a contact force with a tangential component, and therefore does nonzero virtual work.

### C. Euler-Lagrange equation

As D'Alembert principle reformulate Newton's law within the configuration space, it is natural to express it not as a function of Cartesian virtual displacements  $\delta \vec{r}_i$  but generalized virtual displacements  $\delta q^\alpha$  that acts as a basis of the configuration space. The two are related with the chain rule:

$$\delta \vec{r}_i = \sum_{\alpha} \frac{\partial \vec{r}_i}{\partial q^\alpha} \delta q^\alpha \quad (11)$$

D'Alembert principle thus rewrites as

$$\sum_{i,\alpha} \left( m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} - Q_\alpha \right) \delta q^\alpha = 0 \quad (12)$$

With  $Q_\alpha$  the forces expressed with generalized coordinates:

$$Q_\alpha = \sum_i \vec{F}_i^{app} \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} \quad (13)$$

#### Understand the objects

Throughout this chapter, it is important to clearly distinguish the different mathematical objects involved in analytical mechanics.

The **configuration space**  $\mathcal{C}$  is the set of all admissible configurations of the system (not necessarily positions alone! It can involve angles, relative distances, or any other parameter constrained by the geometry of the system). Despite its (very error inducing) name, the configuration space  $\mathcal{C}$  is generally *not* a vector space: it is a  $d$ -dimensional differentiable manifold, most often arising in practice as a submanifold of some ambient  $\mathbb{R}^{n+r}$  (with  $d = n$  the number of free coordinates,  $r$  the number of independent constraints), in the sense developed in the section on submanifolds (see the mathematical tools section). Some configuration spaces (such as  $SO(3)$ , the space of orientations of a rigid body) are most naturally treated as abstract manifolds, without reference to an ambient embedding.

To describe configurations concretely, one introduces a local **chart**: a smooth bijection from an open subset  $U$  of  $\mathcal{C}$  to an open subset of  $\mathbb{R}^d$ :

$$\begin{aligned} \varphi : U \subset \mathcal{C} &\rightarrow \mathbb{R}^d \\ p &\mapsto (q^1(p), \dots, q^d(p)), \end{aligned} \quad (14)$$

The real numbers  $q^1(p), \dots, q^d(p)$  obtained by evaluating  $\varphi$  at a configuration  $p$  are the **generalized coordinates** of  $p$ . they associate to a configuration of the system  $p$  a set of parameters

(real scalars)  $q^\alpha(p)$ , the output of the chart. This tuple of  $d$  parameters are not basis vectors (they can be very different parameters like position or angle) of the possible configurations in the ambient space (which is of dimension  $n + r$  anyway): these are just the number of free parameters needed to represent the accessible configurations in the ambient space.

As the system evolves,  $p = p(t)$  traces a trajectory in  $\mathcal{C}$ , and the generalized coordinates  $q^\alpha(t) = q^\alpha(p(t))$  become functions of time: this is the sense in which  $q^\alpha(t)$  is used throughout the chapter.

A **virtual displacement**  $\delta q$  is, by contrast, an infinitesimal motion compatible with the constraints at fixed time: it is tangent to  $\mathcal{C}$ , and is naturally represented by a tangent vector

$$\delta q \in T_p \mathcal{C} \quad (15)$$

attached to the current configuration  $p \in \mathcal{C}$ . The chart  $\varphi$  induces the coordinate basis  $\left(\partial/\partial q^1|_p, \dots, \partial/\partial q^d|_p\right)$  of  $T_p \mathcal{C}$  (the tangent space near  $p$ ), so that every virtual displacement decomposes as:

$$\delta q = \sum_{\alpha=1}^d \delta q^\alpha \frac{\partial}{\partial q^\alpha} \Big|_p \quad (16)$$

The numbers  $\delta q^\alpha$  are the components of the tangent vector  $\delta q$  in this basis (they are the genuine, coordinate-dependent representatives of an intrinsic, coordinate-free object,  $\delta q \in T_p \mathcal{C}$ ). In the "exercises and examples" section the interested reader may find fully corrected exercises both on submanifolds and analytical mechanics to gain familiarity with these concepts.

Therefore, using the fact that Eq. 12 holds for any virtual displacement  $\delta q^\alpha$ , it is rewritten as:

$$\sum_i m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} = Q_\alpha, \forall \alpha \quad (17)$$

As the Cartesian coordinates depends both on generalized coordinates and time (cf Eq. 4), its total time derivative is obtained with the chain rule formula (we will explicitly write  $\dot{X} = \frac{d}{dt}(X)$  in the few next steps for clarity):

$$\frac{d\vec{r}_i}{dt} = \frac{\partial \vec{r}_i}{\partial t} + \sum_\alpha \frac{\partial \vec{r}_i}{\partial q^\alpha} \frac{\partial q^\alpha}{\partial t} = \frac{\partial \vec{r}_i}{\partial t} + \sum_\alpha \frac{\partial \vec{r}_i}{\partial q^\alpha} \dot{q}^\alpha \quad (18)$$

The second equality comes from the fact that the  $q^\alpha(t)$  have no explicit dependence on other  $q_\beta$ , so all partial derivatives collapse to total derivatives. We'll note them  $\dot{q}_\alpha$ . From this last equation, we obtain by taking partial derivatives with respect to generalized coordinates and velocities the following:

$$\begin{cases} \frac{\partial}{\partial q^\alpha} \left( \frac{d}{dt}(\vec{r}_i) \right) = \frac{\partial \vec{r}_i}{\partial q^\alpha} \\ \frac{\partial}{\partial q^\alpha} \left( \frac{d}{dt}(\vec{r}_i) \right) = \frac{\partial^2 \vec{r}_i}{\partial t \partial q^\alpha} + \sum_\beta \frac{\partial^2 \vec{r}_i}{\partial q^\alpha \partial q^\beta} \dot{q}^\beta = \frac{d}{dt} \left( \frac{\partial \vec{r}_i}{\partial q^\alpha} \right) \end{cases} \quad (19)$$

Let's use these two equations to derive Euler-Lagrange equation from D'Alembert principle ! First, let's rewrite the acceleration term using the product rule of derivatives:

$$\sum_i m_i \frac{d^2}{dt^2}(\vec{r}_i) \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} = \sum_i \left[ \frac{d}{dt} \left( m_i \frac{d\vec{r}_i}{dt} \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} \right) - m_i \frac{d\vec{r}_i}{dt} \cdot \frac{d}{dt} \left( \frac{\partial \vec{r}_i}{\partial q^\alpha} \right) \right] \quad (20)$$

From Eq. 19 we can replace the partial derivative of  $\vec{r}_i$  and  $\dot{\vec{r}}_i$  with respect to  $q^\alpha$ . The acceleration terms rewrites as:

$$\sum_i m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} = \sum_i \left[ \frac{d}{dt} \left( m_i \frac{d\vec{r}_i}{dt} \cdot \frac{\partial}{\partial \dot{q}^\alpha} \left( \frac{d}{dt}(\vec{r}_i) \right) \right) - m_i \frac{d\vec{r}_i}{dt} \cdot \frac{\partial}{\partial q^\alpha} \left( \frac{d}{dt}(\vec{r}_i) \right) \right] \quad (21)$$

Only one last step remain: notice that the velocities  $\frac{d}{dt}(\vec{r}_i)$  are functions of the generalized coordinates  $q^\alpha$  and generalized velocities  $\dot{q}^\alpha$  from eq. 4. As a consequence (using back the  $\frac{dX}{dt} = \dot{X}$ ):

$$\begin{cases} \sum_i m_i \dot{\vec{r}}_i \cdot \frac{\partial}{\partial \dot{q}^\alpha} \dot{\vec{r}}_i = \frac{\partial}{\partial \dot{q}^\alpha} \left( \sum_i \frac{1}{2} m_i \dot{\vec{r}}_i^2 \right) \\ \sum_i m_i \dot{\vec{r}}_i \cdot \frac{\partial}{\partial q^\alpha} \dot{\vec{r}}_i = \frac{\partial}{\partial q^\alpha} \left( \sum_i \frac{1}{2} m_i \dot{\vec{r}}_i^2 \right) \end{cases} \quad (22)$$

This makes the total kinetic energy  $E_c = \sum_i \frac{1}{2} m_i \dot{\vec{r}}_i^2$  appears! Finally, we can inject this new form of the acceleration into the rewritten D'Alembert principle:

$$\frac{d}{dt} \left( \frac{\partial}{\partial \dot{q}^\alpha} (E_c) \right) - \frac{\partial}{\partial q^\alpha} (E_c) = Q_\alpha \quad (23)$$

To finally get Euler-Lagrange equation, we should recall that some applied forces called "conservative" can be explicitly derived from a potential:  $\vec{F}_i^{app} = -\partial_{\vec{r}_i} U$ . For such forces, the generalized expression rewrites as:

$$Q_\alpha = \sum_i \vec{F}_i^{app} \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} = - \sum_i \frac{\partial U}{\partial \vec{r}_i} \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} + \sum_i \vec{F}_i^{app,nc} \cdot \frac{\partial \vec{r}_i}{\partial q^\alpha} = - \frac{\partial U}{\partial q^\alpha} + Q_\alpha^{nc} \quad (24)$$

Where  $\vec{F}_i^{app,nc}$  and  $Q_\alpha^{nc}$  denotes the applied forces and generalized forces that are not conservative and do not derive from a potential. Adding this decomposition in d'Alembert principle finally leads to Euler-Lagrange equation:

$$\boxed{\frac{d}{dt} \left( \frac{\partial}{\partial \dot{q}^\alpha} (\mathcal{L}) \right) - \frac{\partial}{\partial q^\alpha} (\mathcal{L}) = Q_\alpha^{nc}} \quad \text{EULER-LAGRANGE EQUATION} \quad (25)$$

The combination  $E_c - U$  appears naturally because conservative forces contribute a potential term to both sides. We therefore introduced  $\mathcal{L} = E_c - U$  the **Lagrangian** of the system.

### Overview

At this stage, we can take a step back and appreciate what was achieved. Starting from a system of  $N$  particles in  $\mathbb{R}^{3N}$ , subject to  $k$  holonomic constraints and described by  $3N$  coupled vector equations, we arrived at  $d = 3N - k$  scalar equations expressed entirely in the generalized coordinates of  $\mathcal{C}$ . Three things happened simultaneously:

- The constraint forces (i.e. the unknown central difficulty of the Newtonian approach) disappeared entirely by construction, through the geometry of virtual displacements. This is D'Alembert principle.
- The redundant coordinates were eliminated: the  $q^\alpha$  (with  $\alpha \in \{1, 2, \dots, d\}$  the  $d$  generalized coordinates) are independent since  $\mathcal{C}$  being a manifold imposes from the regular value theorem that the Jacobian has rank  $k$  (the number of constraints, called codimension). Thus, no ancillary conditions need to be carried alongside the equations of motion and the computation is much more straight forward.
- The entire dynamics is encoded in a single scalar function  $\mathcal{L} = E_c - U$ : once the Lagrangian is written, the equations of motion follow mechanically by differentiation.



Finally, a few words of comparisons with Newtonian mechanics. The whole purpose of what we did to get Euler-Lagrange equation was to reduce the tediousness of the computation by moving from computations in  $\mathbb{R}^{3N}$  with unknown constraints forces and redundant equations to a cleaner system. While this makes computations much simpler, it comes with the drawback of needing to know the configuration space  $\mathcal{C}$ : what geometry is allowed for a given system. Although this may seem at first difficult, this is already what is done with the forces in the Newtonian approach. Knowing the forces applied on a modeled system and knowing the states possible for this system over time is at the end of the day, the same problem. And, even though the price of analytical mechanics is to properly identify the configuration space, there are other genuinely strong analytical or numerical method we will see that helps to achieve this (Lagrange multipliers, implicit methods...).

Remarkably, this same equation will re-emerge from a completely different starting point: the principle of least action, which recasts the entire problem as the search for a path that extremizes a scalar functional. This equivalence is not a coincidence: it reflects a deep geometric structure that will become central in the transition to quantum mechanics.

### 1.1.3 Properties of Euler-Lagrange equation, action and Variational principle

#### A. About potential and energies in the Euler-Lagrange equation

In the newtonian mechanics chapter, we stated that mechanical energy is only conserved when the potential derived from a force is time and velocity independent. This way, we could show that

$$\frac{dE_c}{dt} + \sum_k \frac{dV_k}{dt} = 0 \quad (26)$$

In the standard case, a force is called conservative when it derives from a position-only potential  $U(q)$  via:

$$Q_\alpha = -\frac{\partial U}{\partial q^\alpha} \quad (27)$$

This works precisely because  $U$  does not depend on  $\dot{q}^\alpha$ , so  $\frac{\partial U}{\partial \dot{q}^\alpha} = 0$  and the Euler-Lagrange equation with  $\mathcal{L} = E_c - U$  correctly reproduces the equations of motion. In our derivation, we assumed that any force that is not derived from a potential which depends only on  $q^\alpha$  is considered in  $Q_\alpha^{nc}$ . However, we can introduce a broader definition of the potential.

The most general form of a potential can take is obtained directly from the structure of the Euler-Lagrange operator:  $U$  is a **generalized potential** of the generalized force  $Q_\alpha$  if:

$$Q_\alpha = \frac{d}{dt} \left( \frac{\partial U}{\partial \dot{q}^\alpha} \right) - \frac{\partial U}{\partial q^\alpha} \quad (28)$$

By construction, we can still introduce  $\mathcal{L} = E_c - U$  with such generalized potential and get the Euler-Lagrange equation unchanged. When  $U$  does not depend on  $\dot{q}^\alpha$ , this reduces to the standard conservative case. The generalized potential is therefore not a new postulate: it is the natural extension of the conservative potential to the full Euler-Lagrange structure. However, as we mentioned, with forces derived from a generalized potential (such as the Lorentz force or the inertial forces, cf the example notes), **mechanical energy is not conserved anymore**.

In the Newtonian mechanics chapter, we defined a physical quantity as conserved if it is an observable (a smooth map from the state space, now defined as  $(q^\alpha, \dot{q}^\alpha)$  to  $\mathbb{R}$ ) whose total time derivative vanishes for all trajectories that are solution of the equations of motion.

A natural candidate for a conserved quantity is obtained by computing the total time derivative of the Lagrangian along a solution. Using the chain rule and the Euler-Lagrange equation :

$$\frac{d\mathcal{L}}{dt} = \sum_{\alpha} \left( \frac{\partial \mathcal{L}}{\partial q^{\alpha}} \dot{q}^{\alpha} + \frac{\partial \mathcal{L}}{\partial \dot{q}^{\alpha}} \ddot{q}^{\alpha} \right) + \frac{\partial \mathcal{L}}{\partial t} = \sum_{\alpha} \frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{q}^{\alpha}} \dot{q}^{\alpha} \right) + \frac{\partial \mathcal{L}}{\partial t} \quad (29)$$

We used the fact that  $\frac{\partial q^{\alpha}}{\partial t} = \frac{dq^{\alpha}}{dt} = \dot{q}^{\alpha}$  (same for the velocity) because they are functions of time only. We can then introduce

$$\boxed{H = \sum_{\alpha} p_{\alpha} \dot{q}^{\alpha} - \mathcal{L}} \quad \text{HAMILTONIAN} \quad (30)$$

Where we introduced for each generalized coordinate  $q_{\alpha}$ , its canonically conjugate variable, the generalized momentum as:

$$p_{\alpha} = \frac{\partial \mathcal{L}}{\partial \dot{q}_{\alpha}} \quad (31)$$

### Generalized momentum and non-conservative force

In Cartesian coordinates with  $\mathcal{L} = \frac{1}{2}m\dot{\vec{r}}^2 - U$ , the generalized momentum immediately gives  $p = m\dot{\vec{r}}$ , the standard linear momentum we defined in the newtonian mechanics notes. With the generalized coordinates however, as  $q_{\alpha}$  can either be a position or an angle,  $p_{\alpha}$  needs not have dimensions of a linear momentum: for an angular coordinate, it has dimensions of angular momentum. The Euler-Lagrange equation rewrites compactly as:

$$\dot{p}_{\alpha} = \frac{\partial \mathcal{L}}{\partial q_{\alpha}} + Q_{\alpha}^{gnc} \quad (32)$$

In particular, if  $\mathcal{L}$  does not depend on  $q_{\alpha}$  (a cyclic coordinate), then  $\dot{p}_{\alpha} = 0$ : the conjugate momentum is conserved. This is the Lagrangian expression of Noether's theorem, which we will make precise shortly.

With this rewriting,  $Q^{gnc}$  denotes the generalized non-conservative forces. These are all forces that cannot be derived from a generalized potential using Eq. 28. For example, notice that a dissipative force:  $F = \sum_{\beta} \gamma_{\alpha\beta} \dot{x}_{\beta}$  with  $\gamma_{\alpha\beta}$  the dissipation tensor is not a generalized conserved force. We will talk more about generalized non-conservative forces after introducing Hamilton equations.

The Hamiltonian satisfies:

$$\frac{dH}{dt} = -\frac{\partial \mathcal{L}}{\partial t} \quad (33)$$

This can be proven by recalling  $p_{\alpha}$  and  $q_{\alpha}$  are functions of time only, and by using the chain rule on the total time differential of  $\mathcal{L}$

This is a fundamental result.  $H$  is an observable (this may sound familiar to readers from quantum physics !): it is a well-defined function on the state space  $(q_{\alpha}, \dot{q}_{\alpha})$ . And it is conserved (in the exact same sense of the Newtonian chapter) if and only if the Lagrangian has no explicit time dependence:

$$\frac{\partial \mathcal{L}}{\partial t} = 0 \implies \frac{dH}{dt} = 0 \quad (34)$$

This is the Lagrangian expression of **energy conservation**, and it already hints at a deep connection: time independence of the Lagrangian is a symmetry, and it produces a conserved quantity. This is a special case of Noether's theorem, which we will state precisely shortly.

## B. Effect of rheonomic constraints

The derivation of Euler-Lagrange assumed implicitly in Eq. 22 that the constraint equations do not depend explicitly on time, i.e.  $f(\vec{r}_1, \dots, \vec{r}_N) = 0$  with no  $t$  dependence. Such constraints are called **scleronomic** (from the Greek *skleros*, rigid). When constraints do depend explicitly on time  $f(\vec{r}_1, \dots, \vec{r}_N, t) = 0$ , they are called **rheonomic** (from the Greek *rheos*, flow).

Let's see what happens with rheonomic constraints. Recall that the Euler-Lagrange equation:

$$\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{q}_\alpha} \right) - \frac{\partial \mathcal{L}}{\partial q_\alpha} = Q_\alpha^{gnc} \quad (35)$$

was derived without assuming anything about the time dependence of constraints: it remains valid in the rheonomic case. What changes is the *structure* of the Lagrangian, and therefore the physical interpretation of the quantities derived from it.

**In the scleronomic case**, the position is function only of the generalized coordinates which are in turn function of time. Therefore, the Cartesian velocities are:

$$\dot{\vec{r}}_i = \sum_\alpha \frac{\partial \vec{r}_i}{\partial q_\alpha} \dot{q}_\alpha \quad (36)$$

so we introduced in the text that :

$$\begin{cases} m_i \dot{\vec{r}}_i \cdot \partial_{q_\alpha} \dot{\vec{r}}_i &= \partial_{q_\alpha} (\frac{1}{2} m_i \dot{\vec{r}}_i^2) \\ m_i \dot{\vec{r}}_i \cdot \partial_{\dot{q}_\alpha} \dot{\vec{r}}_i &= \partial_{\dot{q}_\alpha} (\frac{1}{2} m_i \dot{\vec{r}}_i^2) \end{cases} \quad (37)$$

Allowing us to define  $E_c = \sum_i \frac{1}{2} m_i \dot{\vec{r}}_i^2$ , a purely quadratic form in the generalized velocities  $\dot{q}_\alpha$ .

**In the rheonomic case**, there's an explicit time dependency of the position, leading to the velocity:

$$\dot{\vec{r}}_i = \frac{\partial \vec{r}_i}{\partial t} + \sum_\alpha \frac{\partial \vec{r}_i}{\partial q_\alpha} \dot{q}_\alpha \quad (38)$$

Expanding the velocity square, the kinetic energy decomposes into three contributions:

$$E_c = \underbrace{\sum_{\alpha, \beta} A_{\alpha\beta} \dot{q}_\alpha \dot{q}_\beta}_{T_2 \text{ (quadratic)}} + \underbrace{\sum_\alpha B_\alpha \dot{q}_\alpha}_{T_1 \text{ (linear)}} + \underbrace{C}_{T_0 \text{ (constant)}} \quad (39)$$

where the coefficients are explicitly:

$$A_{\alpha\beta} = \sum_i \frac{m_i}{2} \frac{\partial \vec{r}_i}{\partial q_\alpha} \cdot \frac{\partial \vec{r}_i}{\partial q_\beta}, \quad B_\alpha = \sum_i m_i \frac{\partial \vec{r}_i}{\partial t} \cdot \frac{\partial \vec{r}_i}{\partial q_\alpha}, \quad C = \sum_i \frac{m_i}{2} \left| \frac{\partial \vec{r}_i}{\partial t} \right|^2 \quad (40)$$

The  $T_1$  and  $T_0$  terms vanish identically in the scleronomic case ( $\partial_t \vec{r}_i = 0$ ), recovering the standard purely quadratic form.

Therefore, the rheonomic case keeps the previous definition of the Lagrangian  $\mathcal{L} = E_c - U$  (with  $U$  possibly being a generalized, velocity dependent potential), however now the kinetic energy is not a quadratic form anymore; complexifying the resolution and introducing dissipation.

**Consequence for the Hamiltonian.**

In the scleronomic case,  $E_c$  is purely quadratic in  $\dot{q}_\alpha$  and  $\sum_\alpha p_\alpha \dot{q}_\alpha = 2E_c$ . Hence:

$$H = \sum_\alpha p_\alpha \dot{q}_\alpha - \mathcal{L} = 2E_c - (E_c - U) = E_c + U \quad (41)$$

Recall that despite this equality, the Hamiltonian is not necessarily equal to the mechanical energy, as  $U$  is a generalized potential that may be velocity dependent.

In the rheonomic case, applying the same computation to  $E_c = T_2 + T_1 + T_0$  leads to:

$$\sum_\alpha p_\alpha \dot{q}_\alpha = \sum_\alpha \frac{\partial E_c}{\partial \dot{q}_\alpha} \dot{q}_\alpha = 2T_2 + T_1 \quad (42)$$

by Euler's theorem applied to each homogeneous part separately (we recall that this Euler's theorem is for homogeneous function of degree  $k$  and states that  $x_i \partial f / \partial x_i = k f(x_1, \dots, x_n)$ ). Therefore in the rheonomic case, we get:

$$H = 2T_2 + T_1 - (T_2 + T_1 + T_0) - (-U) = T_2 - T_0 + U \quad (43)$$

The Hamiltonian is *not* equal to  $E_c + U = T_2 + T_1 + T_0 + U$  anymore. It remains the correct conserved quantity when  $\partial_t \mathcal{L} = 0$ , but it no longer admits a simple interpretation as kinetic plus (generalized) potential energy.

The rheonomic case does not break the Euler-Lagrange equation, but it does break two things that are often taken for granted:

- The quadratic structure of  $E_c$  in  $\dot{q}_\alpha$ .
- The identification  $H = E_c + U$  is not true anymore: the Hamiltonian is still conserved when  $\partial_t \mathcal{L} = 0$ , but it equals  $T_2 - T_0 + U$ . Be careful that in any case  $E_c + U$  is not necessarily the mechanical energy since  $U$  is a generalized potential.

A moving constraint injects or removes energy from the system through the  $T_1$  and  $T_0$  terms, which is physically consistent: an externally driven constraint does work on the system, so energy conservation in the naive sense should not be expected.

### C. Action and the variational principle

Given a Lagrangian  $\mathcal{L}(q^\alpha, \dot{q}^\alpha, t)$  and two fixed configurations parametrized by the generalized coordinates  $q^\alpha(t_1)$  and  $q^\alpha(t_2)$ , we define the **action** as the functional:

$$S : \mathcal{T} \rightarrow \mathbb{R} \\ q \mapsto \int_{t_1}^{t_2} \mathcal{L}(q^\alpha(t), \dot{q}^\alpha(t), t) dt \quad (44)$$

$S$  is not a function of time or of the coordinates: it is a map from the set of trajectories (or paths) in the configuration space  $\mathcal{T} = \{q : [t_1, t_2] \subset \mathbb{R} \rightarrow \mathcal{C}\}$ .

#### Hamilton's principle.

Among all paths  $q^\alpha(t)$  connecting two fixed configurations at times  $t_1$  and  $t_2$ , the physically realized trajectory is the one that makes the action **stationary**:

$$\delta S = 0 \quad \text{VARIATIONAL PRINCIPLE} \quad (45)$$

This is **Hamilton's principle** (also called variational principle or the principle of least action, though stationary rather than minimal is the precise term). It is a complete and equivalent reformulation of the Euler-Lagrange equations, derived from a single scalar object rather than from force decompositions.

Blue lines –Least action and paths closeby

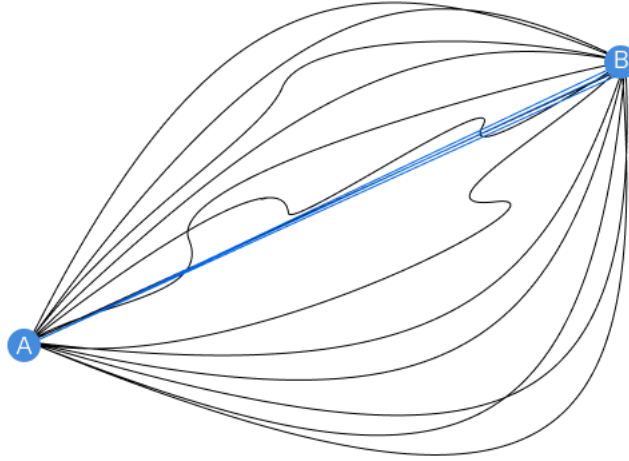


Figure 4: Illustration of the least action principle. Taken from [this blog](#) The blue trajectory is the path chosen by the system. If we disturb the system by considering a trajectory  $q(t) + \epsilon\eta(t)$ , we get closer for arbitrarily small  $\epsilon$  of  $q$  where the action is stationary. In practice, this is why we differentiate the action as in Eq. 46 to look for the stationary path.

#### Derivation of Euler-Lagrange from $\delta S = 0$ .

Consider a family of paths  $q^\alpha(t) + \epsilon\eta^\alpha(t)$ , where  $\eta^\alpha(t)$  is an arbitrary smooth variation vanishing at the endpoints:

$\eta^\alpha(t_1) = \eta^\alpha(t_2) = 0$ . Stationarity of  $S$  requires:

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} S[q + \epsilon\eta] = \int_{t_1}^{t_2} \sum_{\alpha} \left( \frac{\partial \mathcal{L}}{\partial q^\alpha} \eta^\alpha + \frac{\partial \mathcal{L}}{\partial \dot{q}^\alpha} \dot{\eta}^\alpha \right) dt = 0 \quad (46)$$

Integrating the second term by parts and using  $\eta^\alpha(t_1) = \eta^\alpha(t_2) = 0$ :

$$\int_{t_1}^{t_2} \sum_{\alpha} \left( \frac{\partial \mathcal{L}}{\partial q^\alpha} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^\alpha} \right) \eta^\alpha dt = 0 \quad (47)$$

Since this must hold for all variations  $\eta^\alpha$ , the fundamental lemma of calculus of variations gives, for each  $\alpha$ :

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^\alpha} - \frac{\partial \mathcal{L}}{\partial q^\alpha} = 0 \quad (48)$$

This is the Euler-Lagrange equation; recovered here without any reference to forces, virtual displacements, or D'Alembert's principle.

#### Two approaches to the same equation.

It is worth pausing to appreciate what just happened. We now have two completely independent derivations of the Euler-Lagrange equation:

- **D'Alembert's path:** start from Newton's law, eliminate constraint forces via virtual displacements, rewrite in generalized coordinates.
- **Hamilton's path:** postulate that the action is stationary, vary the path, integrate by parts.

Their equivalence reflects a deep consistency between the force-based and variational pictures of mechanics. Hamilton's principle is in many ways the more fundamental of the two: it generalizes immediately to field theory and quantum mechanics, where the notion of force becomes secondary or disappears entirely. **Basically, Hamilton's principle is not built on Newton laws anymore. Only on paths and fields.**

**Remark: stationary, not minimal**

The principle is often called the "principle of least action", but the action is in general only *stationary* (zero first variation), not necessarily minimal. For short time intervals and simple systems it is indeed a minimum, but for longer trajectories or systems with conjugate points, it can be a saddle point. The correct statement is always  $\delta S = 0$ .

## 1.1.4 Summary: Euler-Lagrange equations

- ⊛ **Constraints:** Rather than considering as a state space  $\mathbb{R}^3$  and its tangent space as in Newtonian mechanics, the starting point in analytical mechanics are equations of the shape:

$$f_a(\vec{r}_1, \dots, \vec{r}_N, t) = 0 \quad a \in 1, \dots, k \quad (49)$$

These equations define constraints representing what must physically be enforced (my hand cannot go through the table, the pendulum mass must be at distance  $l$  from the rotation axis...). The above equations are called "holonomic" because they involve the positions only. Time dependent constraints are called rheonomic, time independent constraints are called scleronomic. Unless stated otherwise, we consider here scleronomic constraints.

- ⊛ **Configuration space:** Should we collect the  $k$  above holonomic constraints as a single map  $F = (f_1, \dots, f_k)$ , the physical configurations are given by the level state called "configuration space":

$$\mathcal{C} = F^{-1}(\{0, \dots, 0\}) \quad (50)$$

Despite its name,  $\mathcal{C}$  is not a vector space. If the constraints are all independent (meaning that  $(0, \dots, 0)$  is a regular value of  $\mathcal{C}$ ), the regular value theorem guarantee this is a submanifold. If the ambient space has dimension  $3N$ , then  $\mathcal{C}$  has dimension  $d = 3N - k$ .  $d$  is often called "degrees of freedom" of the system.

- ⊛ **Generalized coordinates:** As a  $d$ -dimensional submanifold,  $\mathcal{C}$  admits locally charts:

$$\begin{aligned} \varphi : U \subset \mathcal{C} &\rightarrow \mathbb{R}^d \\ s &\mapsto (q^1, \dots, q^d) \end{aligned} \quad (51)$$

The  $d$  outputs  $q^d$  of these charts are called "generalized coordinates" and uniquely characterize a configuration in  $\mathcal{C}$ . Through derivation, these coordinates induce a basis of the tangent space near any configurations. This allows to express **generalized velocities**  $\dot{q}^\alpha$ .

- ⊛ **State space:** A complete mechanical state consists of a configuration  $q \in \mathcal{C}$  together with an admissible velocity  $v_q \in T_q\mathcal{C}$ . The natural state space is therefore the tangent bundle:

$$\mathcal{S} = T\mathcal{C} \quad (52)$$

Although  $T\mathcal{C}$  is generally **not a vector space** (again, "state space" was an unfortunate name...), it is a  $2d$ -dimensional manifold. In local generalized coordinates, a state is represented by  $(q^\alpha, \dot{q}^\alpha)$ , giving  $2d$  local coordinates. The dynamics then comes out as a map  $X : \mathcal{S} \rightarrow T\mathcal{S} = T(T\mathcal{C})$ . In comparison, the state space of Newtonian mechanics was  $\mathbb{R}^3 \times T\mathbb{R}^3 \simeq \mathbb{R}^3 \times \mathbb{R}^3$ , which had for a system of  $N$  points dimension  $6N$ . Here, the  $k$  independent holonomic constraints reduce the dimension of the state manifold from  $6N$  to  $6N - 2k$ . This is one of the central interests of the Lagrangian formalism.

- ⊛ **Virtual displacements:** At a fixed time  $t$ , a virtual displacement  $\delta q$  is an infinitesimal variation compatible with the constraints. As such, it is a tangent vector:  $\delta q \in T_q\mathcal{C}$ . It is called "virtual" because it doesn't represent any real evolution (contrary to  $dq = \dot{q}dt$ ): time is fixed.

As a tangent vector, it can be decomposed in the basis of tangent space induced by generalized coordinates:

$$\delta q = \delta q^\alpha \frac{\partial}{\partial q^\alpha} \quad (53)$$

- ⊛ **D'Alembert principle:** For a physically allowed virtual displacement, there is no virtual work induced:

$$\sum_i (F_i - m_i a_i) \cdot \delta r_i = 0 \quad (54)$$

It is crucial to notice that forces meant to represent the constraints automatically vanishes in this equation, because constraints are always orthogonal to the virtual displacement. This is where Newton's equations are projected to a lower dimension state space.

- ⊛ **Euler-Lagrange equation:** Expressing D'Alembert principle in generalized coordinates leads to:

$$\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{q}^\alpha} \right) - \frac{\partial \mathcal{L}}{\partial q^\alpha} = Q_\alpha^{nc} \quad (55)$$

$\mathcal{L}$  is called the **Lagrangian** of the system. In the case of scleronomic constraints,  $\mathcal{L} = E_c - U$ .  $Q_\alpha^{nc}$  are the generalized forces that cannot be derived from the potential.

- ⊛ **Generalized potential and Hamiltonian:** in Newtonian mechanics, we assumed in order to conserve mechanical energy that potential are time and velocity independent. With Euler-Lagrange equations, we don't need the potential to be velocity independent anymore to let the equations unchanged: we can consider forces derived from "generalized potential". Naturally, with these generalized potential, mechanical energy is not conserved anymore. However, the associated conserved energy can be introduced: the Hamiltonian:

$$H = \sum_\alpha p_\alpha \dot{q}^\alpha - \mathcal{L} \quad (56)$$

With  $p_\alpha = \frac{\partial \mathcal{L}}{\partial \dot{q}^\alpha}$  the generalized momentum. As indicated by the lower index,  $p_\alpha$  is a covector (a linear form on  $T_q \mathcal{C}$ ). We also get in the scleronomic case  $H = E_c + U$ . It is important to note that the Hamiltonian is conserved only if the Lagrangian has no explicit time dependency. It is thus still impossible to consider time-dependent potential: this is precisely because energy conservation (whether we consider the mechanical one, or its generalization as a Hamiltonian) requires time invariance symmetry.

- ⊛ **Rheonomic constraints:** We mentioned  $\mathcal{L} = E_c - U$  is only true for scleronomic constraints. But Euler-Lagrange equations still holds with rheonomic constraints. In this case, we can picture the situation as a moving family of configuration space with time. Because of this, the kinetic energy is no longer a nice quadratic form. Instead, it becomes:

$$E_c = \underbrace{\sum_{\alpha, \beta} A_{\alpha\beta} \dot{q}_\alpha \dot{q}_\beta}_{T_2 \text{ (quadratic)}} + \underbrace{\sum_\alpha B_\alpha \dot{q}_\alpha}_{T_1 \text{ (linear)}} + \underbrace{C}_{T_0 \text{ (constant)}} \quad (57)$$

where the coefficients are explicitly:



$$A_{\alpha\beta} = \sum_i \frac{m_i}{2} \frac{\partial \vec{r}_i}{\partial q_\alpha} \cdot \frac{\partial \vec{r}_i}{\partial q_\beta}, \quad B_\alpha = \sum_i m_i \frac{\partial \vec{r}_i}{\partial t} \cdot \frac{\partial \vec{r}_i}{\partial q_\alpha}, \quad C = \sum_i \frac{m_i}{2} \left| \frac{\partial \vec{r}_i}{\partial t} \right|^2 \quad (58)$$

As a consequence, computations are heavier and the Hamiltonian cannot be written as  $E_c + U$  neither.

- ⊛ **Action and variational principle:** given a Lagrangian and two given configurations parametrized by  $q^\alpha(t_1)$  and  $q^\alpha(t_2)$ , we can introduce the action functional:

$$S : \mathcal{T} \rightarrow \mathbb{R} \\ q \mapsto \int_{t_1}^{t_2} \mathcal{L}(q^\alpha(t), \dot{q}^\alpha(t), t) dt \quad (59)$$

The action is thus a map from the set of trajectories in the configuration space  $\mathcal{T} = \{q : [t_1, t_2] \subset \mathbb{R} \rightarrow \mathcal{C}\}$ . The extremum of the action (obtained by vanishing the functional derivative  $\delta S$ ) turns out to be solution of the Euler-Lagrange equation.

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